Shivam Goel

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Website LinkedIn GitHub Google Scholar

Employment

University of Alberta & Amii
Postdoctoral Researcher (incoming) *Starting Dec 2026*

Education

Tufts University Medford, MA
Ph.D. in Computer Science *Aug 2026*

- **Thesis**: Structured Abstractions for Learning and Adaptation in Open World Environments
- **Advisors**: Prof. Jivko Sinapov & Prof. Matthias Scheutz

Washington State University Pullman, WA
Master of Science in Computer Science *May 2017*

- **Thesis**: Engineering a Smart Scarecrow: Bird Deterrence through UAS in Blueberry Fields
- **Advisor**: Prof. Matthew E. Taylor

Publications

* denotes equal contribution.

Conference Publications

1. **Shivam Goel***, Shijie Fang*, Wenchang Gao*, Matthias Scheutz, Jivko Sinapov. FLEX: A Framework for Learning Robot-Agnostic Force-based Skills Involving Sustained Contact Object Manipulation. **ICRA**, 2025.

2. Pierrick Lorang, **Shivam Goel**, Yash Shukla, Patrick Zips, Matthias Scheutz. A Framework for Neurosymbolic Goal-Conditioned Continual Learning for Open World Environments. **IROS**, 2024.

3. James Staley, **Shivam Goel**, Yash Shukla, Elaine Schaertl Short. Agent-Centric Human Demonstrations Train World Models. Reinforcement Learning Conference (**RLC**), 2024.

4. **Shivam Goel**, Yichen Wei, Panagiotis Lymperopoulos, Klara Chura, Matthias Scheutz, Jivko Sinapov. NovelGym: A Flexible Ecosystem for Hybrid Planning and Learning Agents Designed for Open Worlds. **AAMAS**, 2024.

5. Yash Shukla, Bharat Kesari, **Shivam Goel**, Robert Wright, Jivko Sinapov. Few-Shot Policy Transfer through Observation Mapping and Behavior Cloning. **IROS**, 2023.

6. **Shivam Goel***, Yash Shukla*, Vasanth Sarathy, Matthias Scheutz, Jivko Sinapov. RAPid-Learn: A Framework for Learning to Recover for Handling Novelties in Open-World Environments. **ICDL**, 2022.

7. Faizan Muhammad, Vasanth Sarathy, Gyan Tatiya, **Shivam Goel**, Saurav Gyawali, Mateo Guaman, Jivko Sinapov, Matthias Scheutz. A Novelty-Centric Agent Architecture for Changing Worlds. **AAMAS**, 2021.

8. Vasanth Sarathy, Daniel Kasenberg, **Shivam Goel**, Jivko Sinapov, Matthias Scheutz. Spotter: Extending Symbolic Planning Operators Through Targeted Reinforcement Learning. **AAMAS**, 2021.

9. **Shivam Goel**, Santosh Bhusal, Matthew E. Taylor, Manoj Karkee. Detection and Localization of Birds for Bird Deterrence Using UAS. **ASABE** Annual Meeting, 2017.

10. Santosh Bhusal, **Shivam Goel**, Kapil Khanal, Matthew E. Taylor, Manoj Karkee. Bird Detection, Tracking and Counting in Wine Grapes. **ASABE** Annual Meeting, 2017.

Journal Publications

1. **Shivam Goel***, Panagiotis Lymperopoulos*, Ravenna Thielstrom, Evan Krause, Patrick Feeney, Pierrick Lorang, Sarah Schneider, Yichen Wei, Eric Kildebeck, Stephen Goss, Michael C. Hughes, Liping Liu, Jivko Sinapov, Matthias Scheutz. A Neurosymbolic Cognitive Architecture Framework for Handling Novelties in Open Worlds. Artificial Intelligence Journal (**AIJ**), 2024.

2. Garrett Wilson, Christopher Pereyda, Nisha Raghunath, Gabriel de la Cruz, **Shivam Goel**, Sepehr Nesaei, Bryan Minor, Maureen Schmitter-Edgecombe, Matthew E. Taylor, Diane J. Cook. Robot-Enabled Support of Daily Activities in Smart Home Environments. **Cognitive Systems Research**, 2019.

3. Santosh Bhusal, Kapil Khanal, **Shivam Goel**, Manoj Karkee, Matthew E. Taylor. Bird Deterrence in a Vineyard Using an Unmanned Aerial System. Transactions of the ASABE, 2019.

Workshop Publications

1. Pierrick Lorang, **Shivam Goel**, Patrik Zips, Jivko Sinapov, Matthias Scheutz. Speeding-Up Continual Learning Through Information Gains in Novel Experiences. PRL Workshop @ **IJCAI**, 2022.

2. **Shivam Goel***, Gyan Tatiya*, Matthias Scheutz, Jivko Sinapov. NovelGridworlds: A Benchmark Environment for Detecting and Adapting to Novelties in Open Worlds. ALA Workshop @ **AAMAS**, 2021.

Research Experience

Tufts University — Medford, MA

Doctoral Researcher — *Jan 2020 – Aug 2026*

- **Lab affiliations**: Mulip Lab, Jan 2020 – Aug 2026; HRI Lab, May 2023 – May 2026. Advisors: Prof. Jivko Sinapov and Prof. Matthias Scheutz.
- Co-developed FLEX, a framework for learning robot-agnostic, force-based skills for sustained-contact object manipulation (ICRA, 2025); evaluated learned policies in simulation and on physical robots.
- Developed NovelGym, an evaluation ecosystem for hybrid planning and learning agents in open-world environments (AAMAS, 2024).
- Co-developed RAPid-Learn, integrating planning and reinforcement learning to recover from execution failures caused by environmental novelties (ICDL, 2022).
- Integrated learned manipulation policies with symbolic planning in neurosymbolic robotic execution systems; co-authored work on cognitive architectures for novelty handling (Artificial Intelligence, 2024).

Washington State University — Pullman, WA

Researcher — *Oct 2016 – Dec 2019*

- **Lab affiliations**: IRL Lab, Oct 2016 – Dec 2019 (Advisor: Prof. Matthew E. Taylor); CASAS Lab, Jan 2018 – Dec 2019 (Advisor: Prof. Diane J. Cook).
- Developed vision-based bird detection and localization for an autonomous UAV bird-deterrence system in agricultural environments; co-authored ASABE conference and journal publications.
- Integrated object detection and ROS-based deployment on a custom TurtleBot for assistive robotics in smart homes; co-authored a Cognitive Systems Research journal article (2019).

University of Southern California — Los Angeles, CA

Research Intern — *Apr 2018 – Aug 2018*

- **Research affiliation**: Center for Artificial Intelligence in Society (CAIS). Advisors: Prof. Milind Tambe and Prof. Bistra Dilkina.
- Applied AI to wildlife conservation, developing UAV-based thermal vision methods for poacher detection.
- Improved detection accuracy of Faster RCNN models for object detection.

Teaching Experience

Tufts University Medford, MA

Instructor — Introduction to Data Structures *May 2024 – Aug 2024*

- Co-designed and delivered lectures on data structures and algorithms, developed assignments, and evaluated student performance.

Tufts University Medford, MA

Teaching Assistant — Artificial Intelligence (Instructor: Prof. Fabrizio Santini) *Sep 2023 – May 2024*

- Assisted students with AI theory and implementation of learning algorithms.
- Coordinated class assistants for grading and assignment development.

Washington State University Pullman, WA

Teaching Assistant — Operating Systems (Instructor: Prof. K.C. Wang) *Sep 2018 – Dec 2019*

- Co-developed programming assignments in C++ and assisted with grading.

Washington State University Pullman, WA

Teaching Assistant — Software Engineering (Instructor: Prof. David Bakken) *Sep 2017 – May 2018*

- Supported course projects and graded assignments in C#.

Washington State University Pullman, WA

Teaching Assistant — Algorithms (Instructor: Prof. Zhe Dang) *Sep 2016 – Dec 2016*

- Graded assignments and assisted students with algorithmic problem solving.

Research Mentoring

- **Yichen Wei** – Undergraduate researcher, Tufts University; *subsequently M.S. student, Brown University.*
- **Shijie Fang** – Master's researcher, Tufts University; *subsequently Ph.D. student.*
- **Klara Chura** – Undergraduate researcher, Tufts University; *subsequently M.S. student, Brown University.*
- **Turner Hayes** – Master's researcher, Tufts University; *subsequently front-end visualization developer, Boston Dynamics.*
- **Sam Sherman** – Summer intern, Tufts University.
- **Tomas Maranga** – Undergraduate researcher, Tufts University.

Invited Talks

Tufts University Dec 2025

Reinforcement Learning for Force-Based Robot-Agnostic Policies *Host: Prof. Jivko Sinapov*

Brown University Aug 2025

Towards Building Autonomous AI Agents for Open-World Environments *Host: Prof. George Konidaris*

Technical Demonstrations

Tufts Robotics Open House | *2024, 2025*

- Demonstrated neurosymbolic systems for novelty handling on Boston Dynamics Spot and LoCoBot to interdisciplinary audiences.
- Demonstrated voice-command understanding, reasoning, and autonomous robot execution.

MassRobotics | *Boston, 2023*

- Demonstrated VR-based interaction with a TurtleBot.

Tufts Graduate Showcase | *2022*

- Presented multimodal robotic interaction systems to cross-disciplinary audiences.

Technical Expertise

Robotics: Contact-rich manipulation, force/torque learning, sim-to-real transfer, task-and-motion planning
Robots: UR5e, Kinova, Boston Dynamics Spot, LoCoBot (5-DOF), TurtleBot
Simulation: MuJoCo, PyBullet, Gazebo, Robosuite, Isaac Gym
Learning: PyTorch; RL/IL pipelines; OpenAI Gym-style environments
Middleware: ROS/ROS2
Programming: Python, C/C++, Java, MATLAB

Service & Outreach

Program Committee Membership

- AAAI Conference on Artificial Intelligence (AAAI) – 2027.
- International Conference on Automated Planning and Scheduling (ICAPS) – 2024–2026.
- Reinforcement Learning Conference (RLC) – 2026.
- Conference on Robot Learning (CoRL) – 2026.

Technical Reviewer – Journals

- Reinforcement Learning Journal (RLJ).
- Journal of Artificial Intelligence Research (JAIR).
- Artificial Intelligence (AIJ).
- IEEE Robotics and Automation Letters (RA-L).

Technical Reviewer – Conferences

- IEEE International Conference on Robotics and Automation (ICRA) – 2024–2026.
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) – 2023–2026.
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS) – 2022, 2024.
- IEEE International Conference on Development and Learning (ICDL).

Workshop Organization

- Co-organizer, 2nd IJCAI Workshop on Artificial Intelligence for Autonomous Driving (AI4AD), 2022.

Academic & Community Engagement

- Organizer and Coordinator: VLA Reading Group at Tufts (2025), RL Reading Group at Tufts SAIL-ON (2020), RL Reading Group at WSU (2019).
- Seminar Coordinator: IRL Lab seminar series at WSU (2016–2017).
- Volunteer Photographer: Eastern Washington Aspirations in Computing Award Ceremony (2018–2019).

Last updated: November 2026